
When Aggregate Fidelity Misses Intervention Response: A Pre-Fitting Screen for LLM Social Simulation

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Abstract

1 Large social simulations often replace repeated LLM calls with cheaper surrogates,
2 whose validation typically centers on average predictive fidelity. We show a failure
3 this evaluation need not flag: a surrogate can show unremarkable aggregate error
4 while being unable to respond to the intervention that motivates the simulation.
5 We propose a pre-fitting screen that fixes a simulator state, varies one named
6 intervention, measures the stochastic teacher repeatedly, and checks whether the
7 candidate representation distinguishes the paired inputs. In a synthetic campus
8 diffusion study, authority-framed templates yield a mean teacher-stated reshare
9 probability 0.0866 higher than matched hedge-framed templates across 60 selected
10 states, while a 27-feature text-blind representation is identical across each pair. Its
11 response must therefore be zero. This collision also creates a minimum KL regret
12 that fitting cannot remove. Using only the paired teacher means, we estimate this
13 pre-fitting floor at 0.01121 nats. Because KL grows quadratically near equality, a
14 meaningful signed probability shift can still appear modest on an aggregate KL
15 scale. A four-representation comparison confirms the structural zero and shows
16 that capacity, learning, and transfer to unseen wording require separate checks. The
17 audit used 564 LLM calls and cost \$0.20.

18 1 Introduction

19 LLM social simulators use repeated model calls as transition rules for diffusion, cooperation, per-
20 suasion, and collective behavior. At scale, these calls are often replaced with learned decision rules,
21 archetypes, or compiled policies [10, 23, 13, 6]. Once a surrogate is fitted, a natural check is its
22 average agreement with the LLM on ordinary rollout rows. That check answers an important question,
23 but not every question the simulation may ask.

24 A study can depend on a contrast between two otherwise identical social states: two messages,
25 policies, prompts, or source cues. If the surrogate representation omits the manipulated information,
26 both sides become the same input. Training may improve their shared prediction, but it cannot make
27 that one input return two answers. Aggregate fidelity can therefore look ordinary while the response
28 that motivates the study is exactly absent.

29 Let x be a fixed social state, z an intervention, $T(x, z)$ the teacher’s outcome probability, and
30 $\phi(x, z)$ a candidate representation. Any surrogate using that representation has the form $S = g \circ \phi$.
31 *Aggregate fidelity* averages teacher–student agreement over a reference distribution. *Intervention-*
32 *response fidelity* asks whether S preserves the signed contrast $\Delta_T(x) = T(x, z_1) - T(x, z_2)$. If
33 $\phi(x, z_1) = \phi(x, z_2)$, then $\Delta_S(x) = 0$ for every g . Whether this aliasing matters is empirical: the
34 teacher response may be negligible, or it may be central to the study.

This distinction fits a broader view of validation as purpose-specific. A model that produces a plausible network, predicts survey responses, or supports useful human–AI interaction has passed a different test in each case [20, 5, 18, 4]. Those tests are important, but they require a fitted system or an external reference. We ask two earlier questions: before fitting, does the representation contain the information needed for the chosen contrast, and what loss floor follows if it does not?

Related work shows that representations should preserve relevant transition structure [8], that observationally fitted surrogates can misjudge interventions [7], and that distillation need not transfer teacher mechanisms [21]. We turn these ideas into a low-cost capacity screen for a stochastic black-box teacher and natural-language interventions. The screen does not replace later checks against human responses, live interactions, or downstream utility [19, 11].

We did not design the original study to exhibit this failure. A frozen acceptance gate returned FAIL, and its postmortem led to the screen(App. G). We contribute: (i) a pre-fitting quantitative diagnosis in which paired teacher means yield a plug-in estimate of the irreducible aliasing floor, locally quadratic in the signed response for Bernoulli outcomes, (ii) a repeatability-aware, state-clamped screen that can reject an insufficient representation before model choice or optimization, and (iii) a controlled empirical case study showing a structural zero response, followed by a four-representation fit that separates capacity, learning, and wording transfer. The collision implication is immediate, and the divergence identity is established mathematics [2, 14]. Our contribution is their use as a pre-fitting diagnostic of the error that training cannot remove.

2 A pre-fitting capacity screen

The screen asks one practical question: *can this representation express the teacher response that the study needs?* It has four steps.

1. Name the target. Before collection, specify the intervention pair, outcome, direction, and an operational threshold δ . This avoids averaging unrelated interventions when the study depends on one contrast.

2. Build controlled pairs. Draw decision states from a baseline rollout. For each state, hold the agent, network, exposure history, and every non-intervention prompt byte fixed. Change only z . Re-running two trajectories would confound the intervention with the different states created by earlier stochastic choices.

3. Measure the teacher repeatedly. Query both sides enough times to separate the response from call noise, and measure variation directly. A tempting repeatability score is cross-entropy to the call mean. It cannot serve that purpose: because cross-entropy is affine in its first argument,

$$\frac{1}{K} \sum_s \text{CE}(p_{js}, \bar{p}_j) = H(\bar{p}_j).$$

Here p_{js} is a soft target and H is Bernoulli entropy. The score depends only on the mean, so $(.5, .5, .5, .5, .5)$ and $(0, .25, .5, .75, 1)$ receive the same value. We instead used a direct variance rule in a disjoint 200-call pilot to choose $K = 3$ repetitions per side. The pilot did not enter the confirmatory estimate (App. G).

4. Audit the representation. Compare $\phi(x, z_1)$ and $\phi(x, z_2)$. A material teacher response with identical represented inputs rejects the representation. Different inputs only pass a necessary capacity check, and they do not show that fitting will learn or generalize the response. When the inputs collide, the paired teacher means also determine the plug-in loss floor quantified in Sec. 4.

The paired construction follows the logic of controlled behavioral tests [16, 12], but the required response is measured from a stochastic teacher rather than supplied as a linguistic or normative rule. Replication, clamping, and a named contrast are direct repairs to the original gate (App. G). The confirmatory screen used 360 calls. Pilot and smoke tests brought the total to 564 calls and \$0.1965.

79 3 Case study: campus diffusion

80 Our synthetic environment contains 50 student agents in a connected social network. When exposed
 81 to a message, the LLM states probabilities for belief, resharing, and attendance. Only resharing is
 82 sampled, and successful reshares create later exposures, so the LLM is part of the transition kernel
 83 rather than an offline annotator. All personas, relations, and messages are synthetic.

84 We froze 60 scenario-unique decision states, 12 per topic. Within each state we replaced only the
 85 quoted message span with one of 15 frozen topic-by-credibility authority/hedge pairs. Hedging is a
 86 reasonable stress-test dimension because its persuasive effect is documented but context-dependent [3].
 87 Every other prompt byte and all 27 structured features were unchanged. We denote this 27-feature
 88 text-blind input representation by R0 and the surrogate fitted on it as the R0 surrogate.

89 For stated reshare probability, the equal-topic mean contrast was

$$\Delta_{AH} = \frac{1}{60} \sum_j (\bar{p}_{A,j} - \bar{p}_{H,j}) = 0.0866.$$

90 The state-bootstrap interval was [0.0670, 0.1063], the template-cluster interval was [0.0593, 0.1148],
 91 and a two-sided within-state permutation test gave $1/100001 \approx 10^{-5}$. Both interval lower bounds
 92 exceeded the frozen threshold $\delta = 0.03$. Because the paired 27-feature vectors are identical, the
 93 screen rejects that representation and implies $\Delta_S = 0$ for every surrogate that reads only those
 94 features. This is a representation result, not a claim that authority has a calibrated effect on people.
 95 LLM-stated probabilities depend on elicitation and aggregation choices [22].

96 4 A pre-fitting aliasing floor

97 The paired teacher measurements determine a representation-imposed loss floor before any surrogate
 98 is fitted. Let $m_j = (\bar{p}_{A,j} + \bar{p}_{H,j})/2$. Because the R0 representation maps both arms to the same
 99 input, their optimal common Bernoulli prediction is m_j , with minimum regret

$$R_{\text{alias},j} = \frac{1}{2} \text{KL}(\bar{p}_{A,j} \| m_j) + \frac{1}{2} \text{KL}(\bar{p}_{H,j} \| m_j).$$

100 This is the equal-weight Jensen–Shannon divergence. Under a uniform framing prior it is $I(Y; Z |$
 101 $X = j)$; averaging over the empirical anchors gives $I(Y; Z | X)$ and $R_{\text{alias}} = 0.01121$ nats. This
 102 plug-in floor depends only on the paired teacher means and their representation collision, not on a
 103 fitted surrogate or fitting procedure.

104 Why can this floor remain numerically modest? Write $\bar{p}_{A,j} = m_j + \Delta_j/2$ and $\bar{p}_{H,j} = m_j - \Delta_j/2$.
 105 For fixed $m_j \in (0, 1)$,

$$R_{\text{alias},j} = \frac{\Delta_j^2}{8m_j(1-m_j)} + O(\Delta_j^4).$$

106 This follows from Bernoulli curvature [9]. Signed response is first-order in Δ_j , whereas the Bernoulli
 107 KL/JSD penalty is locally second-order. It removes direction, and its coefficient depends on the base
 108 rate. A material signed shift can therefore yield a modest divergence penalty.

109 Now consider the fitted surrogate. Let q_j be R0 surrogate’s out-of-fold prediction, common to both
 110 arms. Average Predictive KL [17] and the Bernoulli Bregman centroid identity give

$$\begin{aligned} R_{\text{total},j} &= \frac{1}{2} \text{KL}(\bar{p}_{A,j} \| q_j) + \frac{1}{2} \text{KL}(\bar{p}_{H,j} \| q_j) \\ &= \underbrace{\text{KL}(m_j \| q_j)}_{R_{\text{fit},j}} + R_{\text{alias},j}. \end{aligned}$$

111 R_{fit} is error relative to the best common prediction and may be reduced by better fitting. R_{alias}
 112 cannot be removed while the arms remain aliased. Averaging over anchors,

$$\underbrace{0.03806}_{R_{\text{total}}} = \underbrace{0.02685}_{R_{\text{fit}}} + \underbrace{0.01121}_{R_{\text{alias}}} \text{ nats.}$$

113 Thus the floor is 29% of observed paired-anchor regret under this weighting, not 29% of behavior or
 114 a causal effect. The percentage depends on q_j ; the 0.01121-nat plug-in floor does not. The plug-in
 115 term includes finite-call noise (App. D).

Finally, R0’s scenario-equal cross-fit KL regret was 0.03622 on ordinary rollout rows and 0.03806 on the paired anchors. These are different evaluation distributions, and we did not run an equivalence test. The comparison is only diagnostic: despite zero response on every anchor, R0’s aggregate regret there was not conspicuously larger. The row-weighted check is similar (ordinary 0.03683, anchors 0.03941, App. D). Aggregate fidelity should not be treated as evidence that a surrogate preserves a named signed response.

5 Fitting checks the diagnosis

The screen establishes R0’s structural zero and estimates its representation-imposed loss floor. It does not predict that a different representation will learn the response. We therefore use fitting as a downstream check, not as another premise of the screen. Four representations share the same rows, scenario-level splits, soft-target objective, and nested cross-fitting. Only the appended input block changes.

Representation	Normal KL (row-weighted)	E_{IR} [95% CI]	Recovery
R0: 27 features	.03683	.0972 [.0757,.1226]	.000
R1: + framing one-hot	.03340	.0713 [.0580,.0887]	.870
R2: + frozen embedding	.03074	.0694 [.0573,.0848]	.813
R3: + PCA-32 embedding	.03082	.0664 [.0549,.0810]	.834

Table 1: The same fitting pipeline with four input representations. Normal KL is row-weighted cross-fit KL on ordinary rollout rows. $E_{\text{IR}} = n^{-1} \sum_j |\Delta_{S,j} - \Delta_{T,j}|$ on 60 anchors. Recovery is $\bar{\Delta}_S / \bar{\Delta}_T$ against a teacher mean of 0.0866.

R0 returns zero contrast on every anchor, exactly as the representation implies. R1 supplies an explicit framing label. R2 supplies a frozen sentence embedding. R3 supplies its 32-dimensional PCA projection. These three recover 81–87% of the mean response. Paired post-hoc intervals show lower E_{IR} than R0, but do not separate R1–R3 (App. C). The result supports a limited conclusion: admitting intervention information changed the fitted response from a structural zero to a nonzero value. It does not show that one encoding is best, or that adding information will always recover a teacher response.

The boundary is equally important. On the prespecified 23-state low-credibility subset where neither message appeared in training, R1–R3 do not improve on R0. Template novelty is nearly confounded with credibility and a smaller teacher effect, so we cannot identify the cause. The correct conclusion is that transfer to unseen wording was not demonstrated. Capacity, learning, and generalization are three separate checks.

6 Limitations and conclusion

The screen is necessary, not sufficient. It can reject an aliased representation, but passing does not establish fitted intervention fidelity, wording generalization, realistic diffusion, or human validity. The 60 anchors are also conditional on a selector later found to over-sample long cascades. Reweighted estimates remain 0.073–0.082, but the direction of population bias is not identified. Authority and hedge templates differ in token count on 55/60 states, so the estimand is a frozen template bundle, not a single-factor human causal effect. The teacher is one pinned LLM configuration (App. A) and the outcome is a stated probability; prompting, sampling, and aggregation are fixed but not varied, so the measured response is specific to this elicitation. Details and downstream validation layers are in the appendix.

Aggregate fidelity and intervention-response fidelity answer different questions. The first measures average prediction on a chosen distribution. The second measures whether a model preserves a chosen change. Neither replaces the other. Before replacing an LLM transition rule, state which social comparisons the simulation must support, measure the teacher on clamped pairs, and reject a representation that aliases a material response. After fitting, test the same pairs and report intervention-response error beside aggregate fidelity.

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A Environment, representation, and fitting

A.1 Synthetic environment

The environment is an undirected connected graph of 50 students. Nodes have school year, major, club membership, socialness, trust, and five topic-interest scores. Edges encode tie strength and shared contexts. Eighty fixed scenarios span five topics and specify three seeds, credibility, urgency, controversy, framing, and message text. On exposure, the LLM returns stated probabilities for believe, reshare, and attend. Only resharing is sampled by a seeded Bernoulli draw. Successful reshares create later exposures.

A.2 Teacher configuration.

All smoke-test, pilot, and confirmatory calls for the pre-fitting screen used the pinned model snapshot gpt-5.4-mini-2026-03-17 with reasoning effort set to none. Temperature was set to 0, and the completion cap was 300 tokens. The API was requested to return a JSON object, which was parsed against a strict four-field schema containing believe, reshare, attend, and reason_code. The same configuration was used across paired arms; within each pair, only the quoted message span differed.

247 A.3 The 27-feature representation

248 The text-blind representation contains five groups:

- 249 • **Exposure and network:** exposure count, summed and maximum tie strength, same-major
250 and same-club exposer counts, source diversity, and diffusion step.
- 251 • **Source stance:** endorsing, neutral, and doubting exposer counts.
- 252 • **Condition:** topic one-hot, urgency, credibility, and controversy.
- 253 • **Persona:** socialness, trust, topic interest, and school year.
- 254 • **Interactions:** trust $\times$ credibility, socialness $\times$ club-event, interest $\times$ urgency, trust $\times$
255 doubting exposers, and interest $\times$ endorsing exposers.

256 It excludes message text, comment text, framing identity, agent identity, and diffusion-mechanism
257 labels. It receives numeric urgency, credibility, and controversy, which the teacher encounters through
258 text. The representation is therefore not deprived of all scenario information. Its relevant omission is
259 the intervention itself.

260 Across the 60 anchors, the $60 \times 2 \times 27$ feature matrix has zero mismatches between arms. The paired
261 prompts have zero differing bytes outside the quoted message span. No feature encodes message
262 length, token count, template index, or another indirect framing cue.

263 A.4 Fitting

264 All four reported representations use the same normal rows, exact soft-target objective, nested
265 scenario-level cross-fitting, and inner-fold selection of the L2 penalty. No anchor state contributes
266 to the fold that produces its own prediction. R0 reproduces the archived predictor to a maximum
267 per-row difference of 2.2×10^{-16} .

268 B Anchor selection and inference

269 B.1 Selection and its audit

270 Sixty scenario-unique states were selected from the baseline rollout, with 12 per topic and prespecified
271 coverage of credibility tier, exposure count, and diffusion step. Ten disjoint states were used for the
272 pilot and ten were held in reserve. No state was dropped or replaced after the freeze.

273 A later audit found that lottery numbers had been assigned to candidate rows subject to one selected
274 row per scenario, while the plan treated scenario as the sampling unit. Longer cascades therefore
275 supplied more lottery entries. Across 400 alternative seeds, candidate-row count correlated with
276 selection frequency at $r = .71$. Selected scenarios averaged 59.6 candidate rows, compared with 38.8
277 for unselected scenarios. Within the selected set, row count correlated with the observed contrast at
278 $r = .50$.

279 $\Delta_{AH} = .0866$ is the correct equal-topic mean for these 60 anchors, and the state bootstrap is
280 conditional on them. The within-selected association does not identify the direction of bias for
281 unqueried scenarios. Reweighted sensitivity estimates range from .073 to .082. Ten comparable pilot
282 states average .0674, but that set is too small for a population comparison. The qualitative screen
283 decision remains above $\delta = .03$, while the estimand must be read as anchor-conditional.

284 B.2 Inference assumptions

285 The fixed-template interval uses 20,000 stratified state-bootstrap samples. The template-cluster
286 interval uses 20,000 resamples of the 15 wording pairs and asks whether the conclusion depends
287 on the chosen templates rather than the chosen states. The permutation test uses 100,000 within-
288 state label permutations and assumes the six calls at a state are exchangeable under the null. The
289 schedule randomized the six condition slots within each state, and the 60 states realize 18 distinct
290 authority/hedge call orders.

291 The operational threshold $\delta = .03$ was set as the rounded first-order equivalent of one additional
 292 expected resharer among the mean 34.54 queried agents per scenario. It is not a claim about final
 293 cascade size.

294 C Representation comparison and wording transfer

295 C.1 Status and metrics

296 The representation comparison was added after the primary teacher experiment. R1–R3 and the
 297 metric hierarchy were Git-committed before fitting. No new LLM calls were made. R2 uses
 298 all-MiniLM-L6-v2 at a pinned revision through the official ONNX artifact. R3 fits PCA on the
 299 training fold and retains 32 dimensions.

300 The primary metric is

$$E_{\text{IR}} = \frac{1}{n} \sum_j |\Delta_{S,j} - \Delta_{T,j}|.$$

301 Mean recovery $\bar{\Delta}_S/\bar{\Delta}_T$ is secondary, and thresholded sign agreement is auxiliary. Uncertainty uses
 302 scenario- and template-cluster bootstraps. A paired 20,000-replicate comparison of representations
 303 was added post hoc and resamples anchors and teacher draws jointly.

304 Relative to R0, the paired reduction in E_{IR} is .0259 [.0087, .0428] for R1, .0277 [.0082, .0475] for
 305 R2, and .0308 [.0112, .0507] for R3. These intervals exclude zero, but the three framing-aware
 306 representations do not separate from one another. Recovery intervals include 1, so the results do not
 307 establish systematic attenuation. Normal-row KL also improves from R0 to R1 by 9.3%. Aggregate
 308 and intervention criteria are not orthogonal, and their improvements are not compared as a ratio
 309 because they are measured on different distributions.

310 C.2 Template overlap and held-out wording

311 Fifteen of the 30 template sides used in the anchor experiment appear verbatim in the normal training
 312 corpus. Table 1 therefore mixes in-template and out-of-template cases for R1–R3. This does not
 313 affect R0’s structural zero: a text-blind representation cannot distinguish seen from unseen wording.

314 States where neither side appears in training number 23 at low credibility, two at mid credibility,
 315 and zero at high credibility. Five more high-credibility states have one side seen. Template novelty
 316 is therefore almost confounded with credibility and effect magnitude. On the prespecified 23-state
 317 low-tier subset, the teacher signed contrast is .0317. Because that denominator is small, we report
 318 E_{IR} rather than recovery: .0506 for R0, .0568 for R1, .0566 for R2, and .0565 for R3. A framing
 319 probe on ten held-out template sides has balanced accuracy .500, which is too underpowered to show
 320 that the embedding lacks framing information.

321 The data do not separate template memorization, weak teacher effect, credibility, and representation
 322 limits. A future study should hold out templates within every credibility tier.

323 D Mathematical and reproducibility details

324 D.1 Exact KL decomposition

325 For one anchor, abbreviate $p_A = \bar{p}_{A,j}$, $p_H = \bar{p}_{H,j}$, and $m = (p_A + p_H)/2$. Direct expansion of
 326 Bernoulli KL gives

$$\begin{aligned} & \frac{1}{2} \text{KL}(\text{Ber}(p_A) \parallel \text{Ber}(q)) + \frac{1}{2} \text{KL}(\text{Ber}(p_H) \parallel \text{Ber}(q)) \\ &= \text{KL}(\text{Ber}(m) \parallel \text{Ber}(q)) + \frac{1}{2} \text{KL}(\text{Ber}(p_A) \parallel \text{Ber}(m)) + \frac{1}{2} \text{KL}(\text{Ber}(p_H) \parallel \text{Ber}(m)). \end{aligned}$$

327 The first term is minimized at $q = m$. The last two terms are the equal-weight Jensen–Shannon
 328 divergence [14]. This is the special-case Bregman bias–variance identity for KL [2]. Averaging over
 329 anchors preserves the equality. Under a uniform binary framing variable Z , the JSD is $I(Y; Z \mid X =$
 330 $j)$; averaging over the empirical anchor distribution gives $I(Y; Z \mid X)$ for that distribution.

331 D.2 Replicate variation

332 Pooling all six teacher calls at an anchor splits replicate JSD into a between-framing and within-
333 framing part. Averaged over anchors,

$$0.01256 \approx 0.01121 + 0.00134 \text{ nats.}$$

334 Thus call noise contributes about one tenth as much as the raw between-framing term. The 0.01121
335 term is a plug-in quantity, not a noise-free latent effect. A permutation null centered at 0.00250 gives
336 the more conservative descriptive value 0.00872.

337 D.3 Quadratic approximation and ordinary rows

338 Evaluating $\Delta_j^2/[8m_j(1 - m_j)]$ state by state gives 0.01101 nats, 1.8% below the exact raw term,
339 with state-level correlation $r = .99996$. Substituting only aggregate means gives 0.00522 because
340 squaring before averaging preserves heterogeneity while squaring the mean does not. Under top-1
341 agreement, only 8/60 anchors have $\bar{p}_A$ and $\bar{p}_H$ on different sides of .5; the remaining 52 probability
342 shifts leave the class label unchanged.

343 The main text compares scenario-equal cross-fit KL regret: .03622 on ordinary rollout rows and
344 .03806 on anchors. As a weighting check, counting the six teacher calls separately gives .03941 on
345 anchors, compared with row-weighted .03683 on ordinary rows. We did not run an equivalence test
346 and do not claim that the evaluation distributions are indistinguishable. The comparison is diagnostic
347 rather than a ranking: the anchor regret is not large enough on its face to make the structural failure
348 obvious.

349 E Heterogeneity

350 Forty-five of 60 anchors have a positive authority-over-hedge contrast, and 39 exceed $\delta = .03$. Topic
351 means are all positive: club event .1144, course material .0794, gossip .0539, internship .1439,
352 and policy change .0414. Effects appear larger at high credibility, with more exposers, and at later
353 diffusion steps, but these subgroup patterns have no independent power and are descriptive only.
354 Heterogeneity explains how the original gate’s average across six framing pairs could fall below a
355 threshold that the strongest named contrast exceeded.

356 F Where the screen sits in prior work

357 Prior work evaluates fitted systems, trains interventionally consistent surrogates, or tests behavior
358 after deployment. Our contribution is earlier: paired teacher measurements test whether a candidate
359 representation can express a named intervention response and, when it cannot, quantify the resulting
360 loss floor before fitting.

361 F.1 Purpose-specific validation

362 Agent-based model validation has long been treated as purpose-specific rather than reducible to one
363 score [20]. Recent LLM work examines different validation targets. Chang et al. compare generated
364 and observed network structure [5]. GEMS evaluates predictions against human responses [18],
365 while ChatBench studies live human–AI interaction [4]. Other work evaluates simulators through the
366 real-user utility of systems trained against them [19], or combines synthetic and limited human survey
367 responses [11]. These tests require generated artifacts, human data, interactions, or downstream
368 systems. Our screen asks an earlier and weaker question: before fitting, does the representation contain
369 the information required for the study’s chosen contrast? It does not replace these later validation
370 layers. Broader discussions of LLM social simulation likewise emphasize iterative, purpose-specific
371 evaluation [1].

372 F.2 Representation and targeted preservation

373 DeepMDP links representation quality to the reward and transition structure that the representation
374 must support [8]. Our connection is conceptual: representation adequacy depends on a specified

375 preservation target. Here the target is chosen by the social study, the teacher is a stochastic black box,
376 and the representation is tested before a replacement is learned.

377 CheckList shows how controlled behavioral tests can reveal failures hidden by held-out accuracy [16].
378 Related work on distillation distinguishes student generalization from fidelity to the teacher’s pre-
379 dictive distribution and shows that the two can diverge [17]. Compression can also leave average
380 performance nearly unchanged while sharply degrading under-represented languages or features [15].
381 These are complementary failure modes. Subgroup degradation may be exposed by a targeted
382 slice; representation aliasing instead forces zero response on a named contrast. Repeated paired
383 measurements also make its irreducible loss floor quantifiable before fitting.

384 **F.3 Interventionally consistent surrogates**

385 The closest work directly studies surrogate behavior under interventions. Dyer et al. use causal
386 abstraction to train interventionally consistent surrogates and show that observationally fitted models
387 can mis-rank policy interventions [7]. Wu et al. define transferred mechanisms through shared
388 invariances under counterfactual changes and study training methods that improve this transfer [21].
389 These are stronger objectives for fitted systems. Our screen has a narrower role: it uses a measured
390 teacher contrast to reject a representation that cannot express that contrast before model choice or
391 optimization. When the paired inputs collide, the same measurements provide a plug-in estimate of
392 the representation-imposed loss floor. The later representation comparison is a downstream check,
393 not a replacement for interventionally consistent training.

394 **F.4 Scaling LLM social agents**

395 Scaling methods make different compromises. Light Society combines full LLMs with distilled
396 surrogates for large social simulations and reports that interaction-level and aggregate rollout criteria
397 need not select the same checkpoint [10]. Zhang et al. distill fast, interpretable decision functions
398 from LLM-driven crowds [23]. GenWorld compiles LLM-derived decision signals into offline lookup
399 policies [13], while LLM archetypes trade individual expressiveness for population scale [6]. These
400 systems motivate checking what a replacement must preserve.

401 Not every scaling method fully replaces the teacher. APS retains the designated LLM as an online
402 transition oracle, queries adaptive prototypes and audits, and propagates their responses to similar
403 agents [24]. It is therefore outside the fixed, trained-replacement setting examined in our case. More
404 generally, the screen applies when researchers propose replacing an LLM transition rule, not to every
405 method for reducing simulation cost.

406 **G The executed gate and the origin of the screen**

407 **G.1 Collections and frozen rule**

408 The original gate used three collections. Eighty normal scenarios, balanced at 16 per topic, produced
409 rollout rows for NLL_{struct} . Forty scaffold items were each queried under five prompt scaffolds,
410 giving 200 records for NLL_{ref} . Eighteen semantic-twin pairs covered six pairwise combinations of
411 authority, social proof, firsthand, and hedge. Twin sides shared the graph, seeds, condition scalars,
412 exposure identities, source stance, comment, and sampled reshare trajectory. Only message text
413 differed.

414 For both binding targets $k \in \{\text{believe, reshare}\}$, the rule required all of

$$G_k = NLL_{\text{struct},k} - NLL_{\text{ref},k} > \max(0.05, 3s_{\text{ref},k}), \quad D_k^{\text{LLM}} \geq 0.10, \quad D_k^{\text{struct}} < 0.02.$$

415 D_k^{LLM} averaged the teacher’s absolute change over the 18 pairs at one call per side. D_k^{struct} was
416 the comparator’s contrast and was identically zero under correct clamping. The rule was executed
417 unchanged on a mini model without reasoning, a bundled reasoning configuration, and a full model
418 without reasoning. The reasoning arm also required default temperature and a larger output allowance,
419 so it is not a single-factor manipulation.

420 The historical gate fit the 27-feature logistic model to ten fixed-seed Bernoulli pseudo-observations
421 per teacher probability with row-wise cross-validation. The analyses in the paper use exact soft

Configuration	Believe G/τ	Reshare G/τ	LLM twins B/R	Joint
Mini, no reasoning	.041/.050	.043/.080	.135/.089	FAIL
Mini, reasoning bundle	.058/.059	.063/.088	.183/.104	FAIL
Full, no reasoning	.029/.050	.046/.074	.129/.086	FAIL

Table 2: The executed gate. FAIL means the frozen evidence rule was not established. It does not mean the environment is compressible or that all distillation would fail.

422 targets and nested scenario-level cross-fitting. Gate verdicts are reported as executed and are not
423 recomputed.

424 G.2 Three aggregation failures

425 The postmortem found three failures with different levels of generality.

426 **Call level.** The repeatability reference averaged $\text{CE}(p_{js}, \bar{p}_j)$ over $K = 5$ scaffold responses, with
427 p_{js} as the soft target. Cross-entropy is affine in its first argument, so

$$\frac{1}{K} \sum_s \text{CE}(p_{js}, \bar{p}_j) = \text{CE}(\bar{p}_j, \bar{p}_j) = H(\bar{p}_j).$$

428 The sequence (.5, .5, .5, .5, .5) and the sequence (0, .25, .5, .75, 1) therefore receive the same value.
429 This is an algebraic identity for any data and any threshold: the statistic measured uncertainty of
430 the mean, not dispersion across calls. Replacing CE with KL restores the discarded entropies and
431 yields a generalized Jensen–Shannon divergence. On the scaffold data, the CE reference was 0.531,
432 whereas the replicate JSD was 0.0146 nats.

433 **Intervention level.** The gate averaged six framing-pair effects although its intended claim was
434 that at least one framing contrast could matter. If pair effects are $d_1, \dots, d_m$, then $\bar{d} \leq \max_i d_i$.
435 The interval between them is non-empty whenever effects are heterogeneous. In this instance the
436 averaged reshare contrast was .0891, authority–hedge was .1182, and the frozen .10 threshold lay
437 between them. Heterogeneity makes such an interval possible. It does not guarantee where any
438 chosen threshold will fall.

439 **Criterion level.** The gate conjoined aggregate compressibility and intervention sensitivity, so either
440 could veto the other. In the baseline, the reshare aggregate margin was .043 against .080. Even if
441 authority–hedge had been the frozen contrast and had passed on believe and reshare, the joint gate
442 would still have failed. This problem depends on the chosen conjunction: unlike the call-level result,
443 it is not a universal identity.

444 These findings motivated three repairs: measure call variation directly, compare clamped states, and
445 name the intervention contrast that the study depends on. The failed gate is design provenance rather
446 than a separate contribution. The call-level identity is stated in the main paper because it directly
447 determines how a repeatability-aware screen must be built.

448 G.3 Provenance

449 For the screen, commits occurred in this order: state selection, primary analysis specification, first
450 pilot call, the $K = 3$ rule and confirmatory schedule, and first confirmatory call. Thresholds therefore
451 preceded the data to which they were applied. The repository has no external timestamp, so we call
452 the ordering prospectively Git-frozen rather than formally preregistered. Secondary analyses are
453 labelled as such. For the earlier gate, raw data were excluded from version control and call logs have
454 no timestamps. Only the fact that thresholds predate analysis code can be verified.